Chapter 4: Ethical and Social Issues in Information Systems

Learning Objectives

After studying this chapter, students should be able to answer the following questions:

- Explain what ethical, social, and political issues are raised by information systems.
- Describe the principles that can be used to guide ethical decisions.
- Explain why modern information systems and the Internet make it difficult to protect individual privacy and intellectual property.
- Explain how information systems have affected the laws on accountability and liability, and the quality of everyday life.

Opening Case: Big Data and Privacy

Today, companies and governments collect a huge amount of data about people. With big data tools, they can study this data to guess how a particular person is likely to behave in the future. This is called predictive modeling. It can be very useful, but it can also hurt people. Just like-

- **Insurance.** An insurance company may charge less to people who do not drive late at night. But poorer people are more likely to work late shifts and to travel far for work, so they may end up paying more. The system looks neutral, but the result can be unfair.
- **Hiring.** Some companies use software to screen job applicants. The software may find that people with shorter commutes stay in a job longer, so it gives a lower score to someone with a long commute. This may be statistically true, but is it fair to a qualified applicant?
- **Targeting vulnerable people.** Data brokers sell reports that point to people with money problems, such as retirees with very little savings. A company selling high-cost loans can then target exactly these people. There are very few rules to stop this.

This is a good example of an ethical dilemma: two groups, each with a good reason. Organizations say big data helps them save money, work better, and even make medical progress. Others say that using this data invades privacy and can harm individuals. As a future manager, you will need to see both sides. This chapter gives you the tools to think about such problems.

4.1 What Ethical, Social, and Political Issues Are Raised by Information Systems?

Ethics refers to the principles of right and wrong that individuals use, as free moral agents, to make choices and guide their behavior. The important words here are "free choice". Ethics is about the decisions people make when they have more than one option.

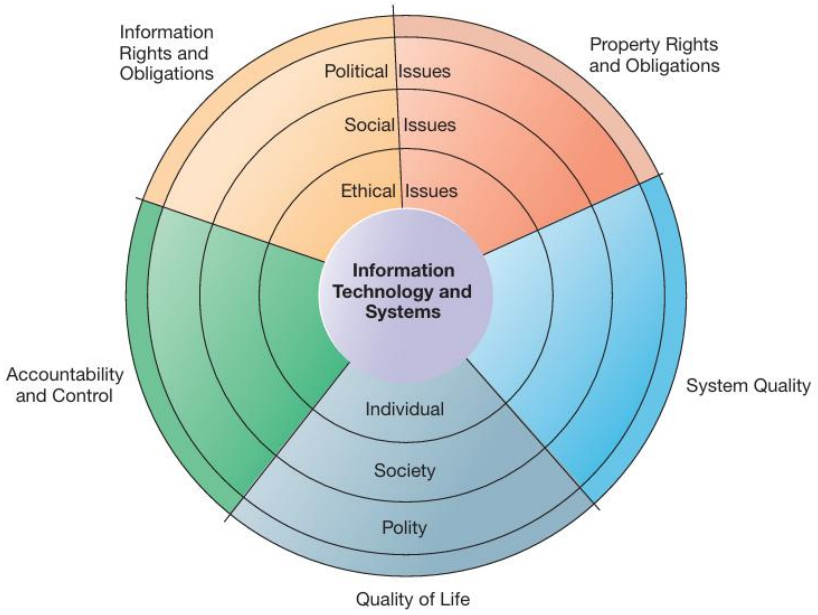
Information systems raise new ethical questions because they create chances for big social change. When a new system arrives, it can shift power, money, rights, and responsibilities from one group to another. Like the steam engine, electricity, the telephone, and radio before it, information technology can bring progress, but it can also be used to commit crimes and harm things society values. It will benefit many people and cost others.

The Internet and e-commerce have made these questions more urgent. It is now very easy to collect, combine, and share information. That raises concerns about how customer information is used, how personal privacy is protected, and how intellectual property is protected.

A Model for Thinking About Ethical, Social, and Political Issues

Ethical, social, and political issues are closely linked. A problem that you face as an individual manager usually shows up later as a debate in society and in politics. The book explains this with a simple picture.

FIGURE 4.1 THE RELATIONSHIP BETWEEN ETHICAL, SOCIAL, AND POLITICAL ISSUES IN AN INFORMATION SOCIETY



Imagine society as a calm pond. People know how to behave in this pond because family, schools, and organizations have developed rules, and laws support those rules. Now throw a rock into the pond. The rock is a new information technology. What happens? Ripples spread out.

- **Individuals** suddenly face new situations that the old rules do not cover.
- **Society** needs time to develop new manners, expectations, and accepted rules. This may take years.
- **Government** also needs time to make new laws, and it often waits to see real harm before it acts.

In the meantime, you may have to act anyway. You may find yourself working in a legal gray area, where there is no clear law to tell you what to do. This is why ethical thinking matters for every manager.

Five Moral Dimensions of the Information Age

The main issues raised by information systems grouped into five moral dimensions.

The five moral dimensions

Moral dimension	Key question	Simple example
Information rights and obligations	What information rights do individuals and organizations have about themselves? What can they protect?	A website tracking what you browse
Property rights and obligations	How can ownership of ideas and creative work be protected when digital copying is so easy?	Pirated software and movies
Accountability and control	Who is responsible and liable when harm is done to people's information or property?	A company whose customer data is hacked
System quality	What level of quality should we demand from data and systems to protect people and society?	Software bugs, cloud outages

Moral dimension	Key question	Simple example
Quality of life	What values and institutions should be protected in an information society?	Job loss, work-life balance, health

Key Technology Trends That Raise Ethical Issues

Ethical issues existed long before computers. But information technology has made them sharper, put pressure on existing social arrangements, and made some laws outdated. Five technology trends are mainly responsible.

Table 4.2 Technology Trends That Raise Ethical Issues

Trend	What it means	Ethical concern
Computing power doubles about every 18 months	More organizations use systems for their core work.	Greater dependence on systems and more damage when systems fail or data is poor.
Data storage costs fall rapidly	It is cheap to keep detailed databases on people.	There is almost no limit on how much data can be kept about you.
Data analysis improves	Companies can study huge amounts of data about individuals.	Detailed personal profiles and large-scale surveillance become possible.
Networking advances	Moving and accessing data becomes cheap and fast from anywhere.	Access to data becomes harder to control.
Growth of mobile devices	Phones are always on and always with us.	A phone can be tracked without the user knowing.

Profiling

Think about all the digital traces you leave: card purchases, phone calls, subscriptions, bank records, government records, and visits to websites. Put together, this data can show far more than

your credit history. It can show your habits, tastes, friends, what you read and watch, and even your political interests. Companies buy such data to target their marketing more precisely.

The use of computers to combine data from many sources and create a detailed digital file on a person is called **profiling**. For example, DoubleClick (owned by Google) tracks visitors across thousands of websites and builds a profile of each visitor's spending and computing habits. This profile is sold to companies for targeted ads. LexisNexis Risk Solutions collects police, motor vehicle, credit, employment, and insurance records to keep files on almost every adult in the United States, and sells them to businesses and government agencies.

Nonobvious Relationship Awareness (NORA)

NORA is a more powerful data analysis technology. It takes data from many different sources, such as job applications, phone records, customer lists, and wanted lists, and finds hidden connections between people. For example, it could discover that a person applying for a job at a casino shares a phone number with a known criminal, and then send an alert to the hiring manager. It can even do this while the data is being created, such as spotting a man at an airline ticket counter who shares a phone number with a known terrorist before he boards the plane. This can help catch criminals, but it also raises serious privacy concerns because ordinary people can be flagged by accident.

4.2 What Specific Principles for Conduct Can Be Used to Guide Ethical Decisions?

Ethics is about individual choice. When you face several courses of action, which one is morally right or an ethical decision?

Basic Concepts: Responsibility, Accountability, and Liability

Concept	Meaning
Responsibility	You are responsible for the costs and consequences of your decisions and must accept the duties and obligations that come with them. This is a key part of ethical behavior.

Concept	Meaning
Accountability	It means we can identify who performed an action and hold that person responsible for it. If we cannot determine who did what, it becomes difficult to ensure ethical behavior and accountability.
Liability	Laws exist that allow people to recover damages done to them by other people, systems, or organizations.
Due process	Laws are known and understood, and a person can appeal to a higher authority to make sure the laws were applied correctly.

Three points follow from these concepts:

1. Information technology does not act by itself. Its effects come from the actions of institutions, organizations, and individuals who use it.
2. The responsibility for the results of technology falls on the institutions, organizations, and managers who choose to use it. If you use technology, you can be held accountable for the results.
3. In an ethical and political society, people can recover damages done to them through laws that follow due process.

Ethical Analysis: A Five-Step Process

When you face a situation that seems to raise an ethical issue, the book suggests following these five steps.

1. **Identify and describe the facts clearly.** Find out who did what to whom, and where, when, and how. Often the first reports are wrong, and just getting the facts straight can help solve the problem.
2. **Define the conflict or dilemma and identify the higher-order values involved.** Every side claims to follow a higher value such as freedom, privacy, protection of property, or free enterprise. A dilemma usually means two opposite actions that both support a

worthwhile value, for example making a company efficient versus respecting individual privacy.

3. **Identify the stakeholders.** Stakeholders are the people and groups who have an interest in the outcome. Find out who they are and what they want.
4. **Identify the options you can reasonably take.** Sometimes no option satisfies everyone, but some options are better than others.
5. **Identify the potential consequences of your options.** An option may be ethically correct but have terrible results in other ways. Ask yourself, "What if I choose this option every time, over a long period?"

Candidate Ethical Principles

After the analysis, which principle should you use to decide? You are the only one who can decide which principles to follow. However, the book lists six principles that have deep roots in many cultures and have lasted throughout history.

Six candidate ethical principles

Principle	What it says	Question to ask yourself
1. The Golden Rule	Do to others what you would want them to do to you.	How would I feel if I were on the receiving end?
2. Kant's categorical imperative	If an action is not right for everyone to take, it is not right for anyone.	If everyone did this, could the organization or society survive?
3. Slippery slope rule	If an action cannot be taken repeatedly, it is not right to take at all. A small, acceptable change today can become an unacceptable change if repeated.	Where will this lead if I keep doing it?
4. Utilitarian principle	Take the action that achieves the higher or greater value.	Which option gives the greater good overall?

Principle	What it says	Question to ask yourself
5. Risk aversion principle	Take the action that causes the least harm or the least potential cost. Avoid actions with very high failure costs.	What is the worst that could happen, and how can I make it less likely?
6. Ethical no-free-lunch rule	Assume that almost everything, tangible or intangible, is owned by someone else unless stated otherwise. If it is useful to you, the creator probably wants payment.	Who made this, and have they been paid or given permission?

Actions that do not easily pass these tests need close attention and great caution. Remember also that even the **appearance** of unethical behavior can harm you and your company as much as real unethical behavior.

4.4 How Have Information Systems Affected Laws for Establishing Accountability and Liability and the Quality of Everyday Life?

New technology also challenges the laws that decide who is accountable and liable. If a person is hurt by a machine that is partly controlled by software, who is responsible? Should a social network be held liable for harmful posts by its users? If you move your data processing to the cloud and the cloud provider fails, what can you do?

Computer-Related Liability Problems

In late 2013, hackers stole credit card, debit card, and personal information of 70 to 110 million customers of Target, a large U.S. retailer. Target spent over $60 million to strengthen its systems, and in 2015 agreed to pay $10 million to customers and $19 million to MasterCard. It also lost sales and customer trust. Who should pay for the harm? Target, which did make some effort to protect the data? Or is it simply a cost of doing business in a credit card world? Note that under U.S. banking law, a customer's maximum liability for credit card theft is $50.

Can software makers be held liable? The book explains it this way:

- When software is part of a **machine** that injures someone, the producer and the operator can be held liable.
- When software works like a **book**, storing and showing information, courts are very reluctant to hold the author or publisher liable. This is partly to protect freedom of expression. So it is very hard to hold software producers liable for software treated like a book.

But software is different from a book in important ways. Users may expect it to be perfect, it is harder to inspect and compare, it performs a task instead of just describing one, and people depend on it every day. Because software is so central to daily life, the book says liability law will probably extend to software in the future, even when it only provides information.

System Quality: Data Quality and System Errors

On December 24, 2012, millions of Netflix customers lost service for a day because Amazon's cloud service (AWS) failed. Later outages also hit services such as Microsoft Azure, Twitter, and Salesforce. This raises the question: what is an acceptable level of system quality? At what point should managers say, "Stop testing, let us go live"?

The difficulty is that some errors can only be fixed at a very high cost. Software companies knowingly release software with small bugs, because fixing every minor error would take so long and cost so much that the product would never be released. Should a company then withdraw the product, warn the user, or leave it to "let the buyer beware"? Individuals and organizations can be held responsible for errors that were foreseeable and avoidable.

Principal sources of poor system performance

Source of poor system quality	Explanation
1. Software bugs and errors	Software of any real complexity cannot be made completely free of defects, and the seriousness of remaining bugs cannot be known in advance.
2. Hardware or facility failures	Caused by natural disasters or other events.

Source of poor system quality	Explanation
3. Poor input data quality	Wrong, incomplete, or outdated data entered into the system.

Quality of Life: Equity, Access, and Boundaries

As technology gets more powerful, its negative social costs are also growing. Many of these harms are not crimes and do not violate anyone's legal rights, but they can still hurt people, society, and political institutions.

Quality of life issues

Issue	What the book says
Balancing power: center versus periphery	People once feared that big central computers would give governments total control, as in Orwell's novel 1984. Decentralized computing reduced this fear. But today a few large Internet companies (Google, Apple, Amazon, Microsoft) collect much of the world's personal data, and governments use this data for surveillance. Power has become more centralized in a few large companies and government agencies.
Rapidity of change	Information systems have made global markets very efficient, so businesses have less time to respond to competitors. A company can be wiped out within a year, along with your job. On-demand services such as Uber create "just-in-time" jobs with no benefits or insurance.
Maintaining boundaries: family, work, and leisure	Smartphones and mobile computing mean we can work anywhere, so work spreads into family time, vacations, and leisure. Heavy Internet and phone use also takes people away from family and friends. Among teenagers, it can lead to harmful behavior such as cyberbullying.

Issue	What the book says
Dependence and vulnerability	Businesses, governments, schools, hospitals, and utilities depend heavily on information systems, so they are highly vulnerable when the systems fail. Unlike the phone or electricity, there are no regulatory or standard-setting bodies for information systems.
Employment	Business process reengineering has cost millions of factory and clerical workers their jobs. Some economists warn that automation now threatens white-collar jobs as well, such as tech support, call center work, X-ray examination, and legal document review. Others argue that displaced workers will find better jobs in growing industries. The groups that may find it hardest to adapt are unskilled workers and older, less educated middle managers.
Equity and access: the digital divide	Computers, knowledge, and Internet access are not shared equally across social classes. Higher-income families are more likely to have computers and broadband, and their children more often use the Internet for learning, while lower-income children more often use it for entertainment. Left uncorrected, this **digital divide** could create a society of information "haves" and "have-nots".

Computer Crime and Computer Abuse

Term	Meaning
Computer crime	Committing illegal acts by using a computer or against a computer system. Even accessing a system without permission, or with intent to harm, even by accident, is now a federal crime in the U.S. Common types are malware, phishing, network interruption, spyware, and denial of service attacks.

Term	Meaning
Computer abuse	Acts involving a computer that may not be illegal but are considered unethical. The best-known example is spamming.

Spam is junk e-mail sent in bulk to people who never asked for it. It is common because it costs only a few cents to send thousands of messages.

Health Risks

- **Repetitive stress injury (RSI)** happens when muscle groups are forced through the same movement again and again. Computer keyboards are the biggest single cause. It may affect up to one-third of the workforce. The most common type is **carpal tunnel syndrome (CTS)**, where pressure on a nerve in the wrist causes pain, numbness, and tingling. RSI can be avoided with better workstation design (wrist rests, proper monitor stands, ergonomic keyboards), frequent rest breaks, and rotating employees between jobs.
- **Computer vision syndrome (CVS)** is eyestrain from using screens such as computers, laptops, e-readers, and smartphones. It affects about 90 percent of people who spend three or more hours a day at a computer. Symptoms include headaches, blurred vision, and dry, irritated eyes, and they are usually temporary.
- **Effects on thinking.** Some experts worry that the Internet makes it harder for people to focus and think clearly, and that looking up answers replaces real problem solving.

Review Questions

1. What ethical, social, and political issues are raised by information systems? Explain how they are connected and give some examples.
2. List and describe the key technology trends that increase ethical concerns.
3. Define profiling and nonobvious relationship awareness (NORA) and explain the ethical concerns they raise.
4. Differentiate between responsibility, accountability, and liability.

5. List and describe the five steps in an ethical analysis.
6. Identify and describe the six ethical principles.
7. Explain why it is so difficult to hold software services liable for failure or injury.
8. List and describe the principal causes of system quality problems.
9. Define computer crime and computer abuse and describe their impacts.
10. Define computer vision syndrome and repetitive stress injury (RSI) and explain how they relate to information technology.
11. Discuss the pros and cons of allowing companies to collect personal data for behavioral targeting.